

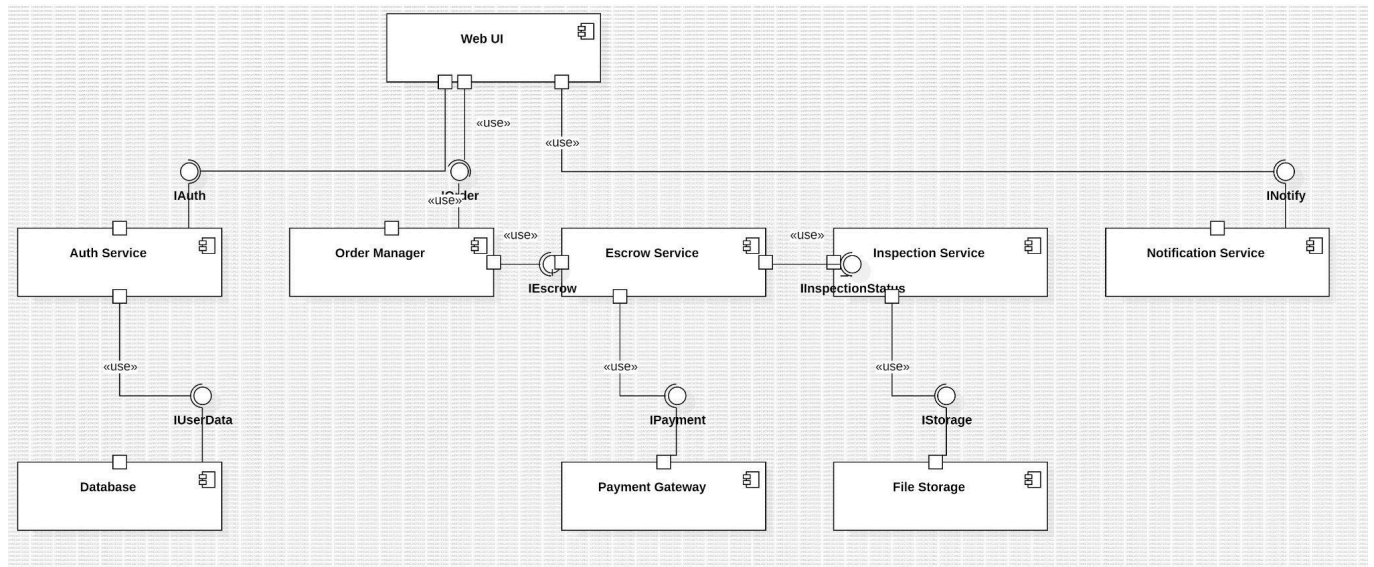
Architectural Style Justification

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COMPONENT DIAGRAM :



Architectural Choice

The system is organized into four layers, a presentation layer holding the web UI, a business layer holding the Auth Service, Order Manager, Escrow Service, Inspection Service and Notification Service, a data layer holding the database, and a set of external systems for payment and file storage. Each layer only talks to the layer directly below it through clearly defined interfaces, so the business rules never leak into the UI and the UI never touches the database directly.

Reason One - Matches the Workflow

The core of this system is a linear order lifecycle, an item is listed, purchased, inspected and then either completed or disputed. A layered structure mirrors that flow naturally, requests move down from the UI through the business logic into the data layer and back up again, which keeps the escrow release logic easy to trace and easy to test as a single, coherent path rather than something spread across independent services.

Reason Two - Right Sized for the Team and Timeline

Microservices would add real operational overhead, separate deployments, service discovery and network calls between every module, none of which a one person team building this in a few days actually needs. A layered monolith still keeps the Auth, Order, Escrow, Inspection and Notification components logically separate and independently testable, but they ship as one deployable unit, which fits both the size of the team and the short build window.

Security Advantage

Because every request must pass through the business layer before it reaches the data layer, there is a single, consistent place to enforce authentication and authorization. The UI never has direct database access, so a compromised or buggy client cannot bypass the Auth Service or reach into escrow records directly, all sensitive operations such as fund release are forced through the Escrow Service's own validation.

Performance Benefit

With everything running inside one application process, calls between the Order Manager, Escrow Service and Inspection Service are simple in process function calls rather than network requests, so there is no added latency from service to service communication. This keeps common actions such as approving a product and releasing escrow funds fast and predictable, which matters more here than horizontal scalability given the expected load.